MIDDLE EAST TECHNICAL UNIVERSITY
Department of Electrical and Electronics Engineering

EE 301 Signals & Systems
Homework 4
Due: December 17, 2017

Problem 1:

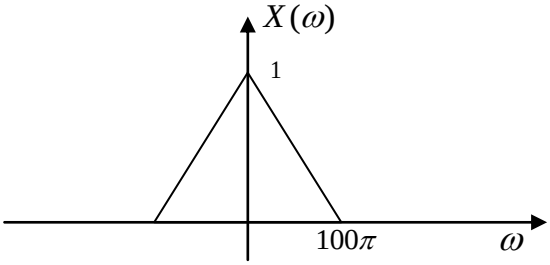
Consider an LTI system with the impulse response $h(t) = \dfrac{\sin(4(t-1))}{\pi(t-1)}$.

Determine the outputs $y_1(t)$, $y_2(t)$ and $y_3(t)$ of this system to the inputs

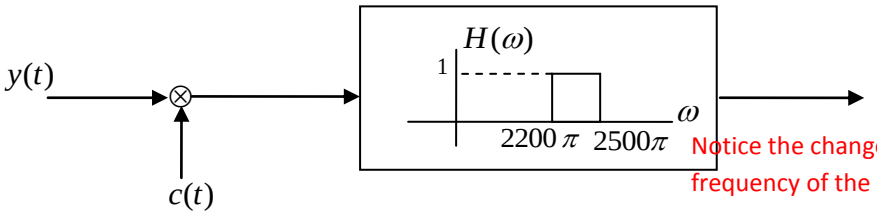
$$x_1(t) = \cos\left(6t + \frac{\pi}{2}\right), \qquad x_2(t) = \frac{\sin(4(t+1))}{\pi(t+1)} \qquad \text{and} \qquad x_3(t) = \left(\frac{\sin 2t}{\pi t}\right)^2.$$

Problem 2:

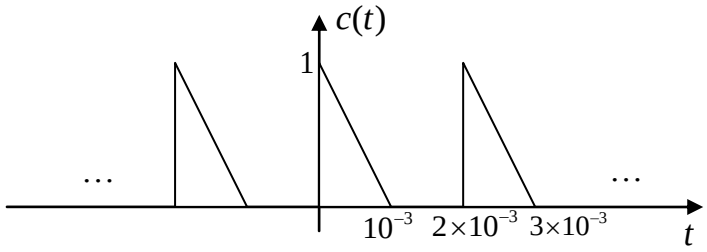
Consider a real and even signal $x(t)$ with a triangular spectrum (Fourier transform) given.



a) Find and draw the spectrum of $y(t) = x(t)\cos(400\pi t)$.
b) Find and draw the spectrum for filter input $b(t)$ and filter output $z(t)$ in the following system, where the filter is an ideal bandpass filter with a real impulse response. (Do not evaluate the FS coefficients of $c(t)$).
c) Express $Z(\omega)$ in terms of $X(\omega)$ and FS coefficients of $c(t)$. Similarly, express $z(t)$ in terms of $x(t)$.



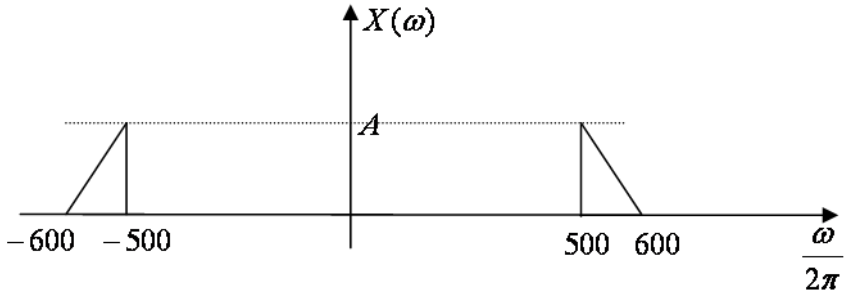
Notice the change in the upper cut-off frequency of the BPF.



Problem 3:

We know that it is possible to reconstruct $x(t)$ from its samples if there is no overlap of spectrum replicas. Nyquist theorem gives a guaranteed condition for perfect reconstruction; but for band-pass signals lower sampling rates can be possible. We examine sampling of band-pass signals (band-pass sampling) in this problem.

The signal $x(t)$ with the spectrum $X(\omega)$ is sampled at various sampling rates. (Assume that the signal is sampled with an ideal impulse train.) Sketch the spectrum of the sampled signal for the sampling rates 300, 1100 and 1200 Hz. For which cases, is the signal reconstructable from its samples by using an ideal BPF?



Problem 4:

A linear, time-invariant and stable system is characterized by the transfer (system) function $H(s) = \dfrac{3+s}{(2+s)^2}$.

a) Determine the ROC for $H(s)$ by explaining your reasoning and find the impulse response $h(t)$.

b) If the system is excited by an input $x(t) = \delta(t) + 2e^{-t}u(t) + te^{-t}u(t)$, find the output $y(t)$ using transform domain techniques.

Problem 5:

Evaluate the time-domain signal $x(t)$ with the Laplace transform

$$X(s) = \frac{s^2}{(s+1)(s^2+4s+5)}$$

if the ROC R_X is specified as

a. $R_X : \Re\{s\} < -2$.
b. $R_X : -2 < \Re\{s\} < -1$.
c. $R_X : -1 < \Re\{s\}$.

Also sketch the pole-zero diagram of $X(s)$. If $X(s)$ were the transfer function (or the system function) of an LTI system, which of the above ROC's would result in stable and/or causal systems?

Problem 6: The relationship between the input voltage $x(t)$ and the output voltage $y(t)$ of an RLC series circuit is given by the following differential equation

$$\frac{d^2y(t)}{dt^2}+\frac{R}{L}\frac{dy(t)}{dt}+\frac{1}{LC}y(t)=\frac{1}{LC}x(t).$$

By determining the poles and zeros of the s-domain transfer function describing the RLC circuit, show that the causal implementation of the RLC circuit is always stable for positive values of R, L, C of the passive components.

Problem 7:

a) Consider an LTI system (System 1) with input $x[n]$ and output $y[n]$ for which

$$y[n]-\frac{5}{2}y[n-1]+y[n-2]=x[n].$$

Find the system function, plot its poles and zeros. Also find the causal impulse response of this system.

b) The input $x[n]$ and output $y[n]$ for an LTI system (System 2) are given as

$$x[n]=(1/3)^n u[n]+2^n u[-n-1], \quad \text{and} \quad y[n]=5(1/3)^n u[n]-5(2/3)^n u[n].$$

Write the difference equation for System 2 using transform techniques.

Problem 8: Find the z-transforms (and the corresponding ROC's) of the following sequences by using the z-transforms of $a^n u[n]$ and $-a^n u[-n+1]$ and required z-transform properties. Also state the related property.

a) $(5/6)^n u[n-6]$
b) $n\,u[n]$
c) $n\,u[-n]$

Please solve and submit all 8 questions. 5 of them will be chosen randomly and evaluated in detail. Remaining 3 questions will be graded on/off, considering the final result.